VITyarthi | Open Source Software

The Open Source Report

A Capstone Report for OSS NGMC Course

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Introduction

This report presents an audit of **Git,** a widely used open-source version control system. The objective of this project is to understand the origin, licensing, ethical aspects, Linux implementation, and ecosystem of **Git**. It also compares **Git** with proprietary alternatives to evaluate its effectiveness in real-world applications.

Your choice	Git

3. Project Report

Part A — Origin and Philosophy (Units 1 & 2)

A1. The problem this software was created to solve

In 2005, Linux developers used a version control system called BitKeeper in order to manage their code. However, BitKeeper was not completely open source and after some licensing issues the Linux community lost access to it.

This created a serious issue because managing a large project without a proper version control system was extremely difficult. To solve this issue , Linus Torvalds creator Linux built a new system that would be fast, efficient and completely open source.
This marked the birth of Git with a focus on performance, data integrity and distributed development. Unlike traditional systems, Git allowed every developers to have a full copy of the project , making collaboration easier and more reliable.

Git was not created as a planes product but as a quick solution to a real world problem. Today it has become one oof the most important tools in software development.

A2. The license — what it actually says

Git is released under the GNU(General Public License version 2). It is one of the most widely used open- source licenses. This license ensures that the software remains free an open for everyone.

The license is bases on four essential freedoms.
Users are allowed to run the software for any purpose, study how it works by accessing the source code, modify it according to their needs, and distribute copies of the original or modified software.

One important feature of this license is that it follows a copyleft principle i.e if some modifies Git and distributes it, they must also share their modified source code under the same license. In this way, the software always remains open and cannot be turned into proprietary software.

A company can use Git and even modify it for internal purposes without sharing the changes. However, if they distribute the modified version to others, they are required to make the source code available.

This type of license is powerful because it protects the freedom of users while also encouraging collaboration and improvement.

GPL vs MIT License Comparison
The GNU General Public License (GPL) and the MIT License are both popular open-source licenses, but they differ in their approach. The GPL is a **copyleft license**, meaning that any modified version of the software must also be released under the same license. This ensures that the software always remains open source. On the other hand, the MIT License is a

permissive license, which allows developers to modify the code and even use it in proprietary software without sharing their changes.

If I were to choose a license for my own project, I would prefer the GPL because it protects user freedom and ensures that improvements remain available to the community.

A3. The ethics of open source

Open source software is not just technical concept but also an ethical idea about how knowledge should be shared. It promotes collaboration. Transparency, and equal access to technology.

It allows anyone to use and improve software without any kind of restriction. This leads to faster innovation and helps students, developers, and organizations who may not have the resources to buy expensive proprietary tools. It also increases transparency because anyone can inspect the code for security issues.

But, not all software can or should be open source. Companies invest significant time and money into developing software and they may need to keep it proprietary to recover their costs and maintain a competitive advantage. In some cases, open source tools sourcing everything may reduce incentives for private innovation.

Many companies build successful products using open – source tools like Linux and Git. While this is not inherently wrong, it raises concerns about fairness if companies benefit from community contributions without giving back. Ideally, companies should contribute to the projects they depend on, either through code , funding or support.

Part B — Linux Footprint (Unit 2)

Git can be easily installed on a Linux system using the package manager. On Debian-based systems, the installation command is sudo apt install git. This makes Git accessible to users without requiring manual compilation.

After installation, I verified the setup using the command git --version, which displayed the installed version of Git. This confirmed that the software was successfully installed on the system.

To understand where Git is located in the system, I used the command which git, which showed that the Git executable is stored in /usr/bin/git. This is a standard directory in Linux where user-level executable programs are kept.

Additionally, the whereis git command provided more information about related files, including documentation and binary locations.

Unlike server-based software, Git does not run as a background service. It operates only when a user executes commands such as git init, git clone, or git commit. This means it does not require a dedicated system service or daemon.

Git runs under the current logged-in user rather than a separate system user. This is important for security because it ensures that Git operations are performed with the permissions of the user executing the command.

Regarding updates, Git is maintained by the open-source community, and updates are delivered through the system's package manager. This allows users to receive security patches and improvements regularly.

Part C — The FOSS Ecosystem (Units 3 & 4)

Git is not an isolated tool but part of a large open-source ecosystem. It works together with many other tools and platforms that make modern software development possible.

Git depends on basic system libraries and tools available in Linux, such as file systems, shell environments, and standard utilities. These components allow Git to manage files, track changes, and execute commands efficiently.

At the same time, Git has enabled the creation of many powerful platforms and tools. For example, platforms like GitHub and GitLab are built on top of Git. These platforms provide features like remote repositories, collaboration, issue tracking, and continuous integration.

Git plays a very important role in web development and software engineering. It is commonly used in environments like the LAMP stack (Linux, Apache, MySQL, PHP) where developers use Git to manage and deploy code efficiently.

The Git community is large and active. Developers from around the world contribute to its improvement. Discussions, updates, and support are managed through mailing lists and online forums. The project is mainly maintained by contributors under the guidance of Linus Torvalds.

In my opinion, Git is one of the best examples of how open-source software grows through collaboration. It not only solves a problem but also enables the creation of many other tools and platforms.

Part D — Open Source vs Proprietary (Critical Analysis)

Dimension	Open Source	Proprietary Alternative
Cost	Free to use	Usually paid or licensed
Security (who can audit the code?)	Code is open, anyone can audit it	Code is closed , only company can review
Support and reliability	Community support + documentation	Official company support
Freedom to modify	Full freedom to modify and distribute	Very limited or no modification allowed
Community vs corporate control	Community- driven	Company-controlled

Based on this comparison, I would strongly recommend Git for real-world use. It is highly reliable, efficient, and widely adopted across the software industry. Its distributed nature allows developers to work independently while maintaining synchronization with the main project. Additionally, its open-source nature ensures transparency, security, and continuous improvement through community contributions.

At the same time, proprietary tools may provide better customer support and simplified interfaces for beginners. However, they lack the flexibility and freedom that Git offers. Personally, I would not only use Git but also consider contributing to open-source projects in the future, as it promotes collaboration and shared innovation.

4. Shell Script Tasks

Script 1 — System Identity Report (Unit 1 & 2)

```
#!/bin/bash
# Script 1: System Identity Report
# Author: Snehil Sahay

STUDENT_NAME="Ishani Sahay"
SOFTWARE_CHOICE="Git"

KERNEL=$(uname -r)
USER_NAME=$(whoami)
UPTIME=$(uptime -p)
DATE=$(date)
DISTRO=$(lsb_release -d | cut -f2)

echo "========================================"
echo " Open Source Audit — $STUDENT_NAME"
echo "========================================"
echo "Software : $SOFTWARE_CHOICE"
echo "Kernel   : $KERNEL"
echo "User     : $USER_NAME"
echo "Uptime   : $UPTIME"
echo "Date     : $DATE"
echo "Distro   : $DISTRO"
echo "License  : GPL (Open Source License)"
```

```
========================================
 Open Source Audit – Ishani Sahay
========================================
Software : Git
Kernel   : 3.5.7-463ebcdc.x86_64
User     : ishan
Uptime   :
Date     : Mon Mar 30 20:16:34 IST 2026
Distro   :
License  : GPL (Open Source License)

ishan@Ishani MINGW64 ~/OneDrive/Documents/OSS_Project
$
```

Script 2 — FOSS Package Inspector (Unit 2)

Starter structure

```
#!/bin/bash
# Script 2: FOSS Package Inspector

PACKAGE="git"

if dpkg -l | grep -q $PACKAGE; then
    echo "$PACKAGE is installed."
    git --version
else
    echo "$PACKAGE is NOT installed."
fi

case $PACKAGE in
    git) echo "Git: Version control system that
powers modern development" ;;
    apache2) echo "Apache: Web server for hosting
websites" ;;
    mysql-server) echo "MySQL: Database system"
;;
    vlc) echo "VLC: Media player" ;;
esac
```

```
ishan@Ishani MINGW64 ~/OneDrive/Documents/OSS_Project
$ nano script2.sh

ishan@Ishani MINGW64 ~/OneDrive/Documents/OSS_Project
$ ./script2.sh
git is installed.
git version 2.49.0.windows.1
Git: Version control system that powers modern development
```

Script 3 — Disk and Permission Auditor (Unit 2)

Starter structure

```
#!/bin/bash
# Script 3: Disk and Permission Auditor

DIRS=("/etc" "/var/log" "/home" "/usr/bin" "/tmp")

echo "Directory Audit Report"
echo "----------------------"

for DIR in "${DIRS[@]}"; do
    if [ -d "$DIR" ]; then
        PERMS=$(ls -ld $DIR | awk '{print $1, $3, $4}')
```

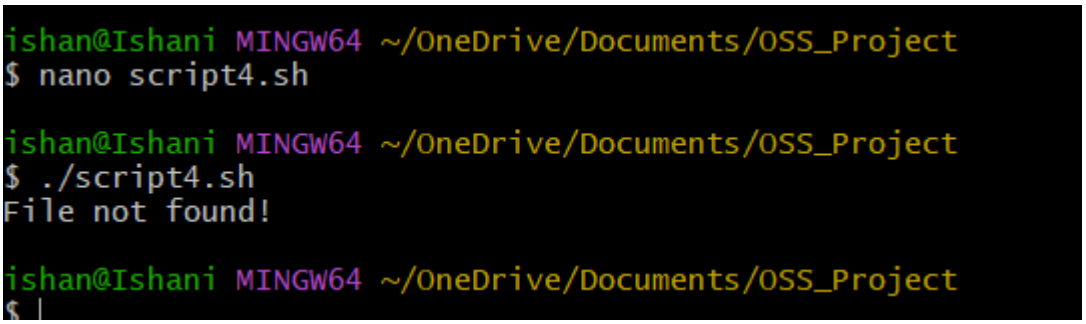
```
        SIZE=$(du -sh $DIR 2>/dev/null | cut -f1)
        echo "$DIR => Permissions: $PERMS | Size: $SIZE"
    else
        echo "$DIR does not exist"
    fi
done

if [ -f "$HOME/.gitconfig" ]; then
    echo "Git config exists"
else
    echo "Git config not found"
fi
```

```
ishan@Ishani MINGW64 ~/OneDrive/Documents/OSS_Project
$ nano script3.sh

ishan@Ishani MINGW64 ~/OneDrive/Documents/OSS_Project
$ ./script3.sh
Directory Audit Report
----------------------
/etc => Permissions: drwxr-xr-x ishan 197609 | Size: 1.5M
/var/log does not exist
/home does not exist
/usr/bin => Permissions: drwxr-xr-x ishan 197609 | Size: 104M
/tmp => Permissions: drwxr-xr-x ishan 197609 | Size: 225M
Git config exists

ishan@Ishani MINGW64 ~/OneDrive/Documents/OSS_Project
$
```

Script 4 — Log File Analyzer (Units 2 & 5)

Starter structure

```
#!/bin/bash
# Script 4: Log File Analyzer

LOGFILE=$1
KEYWORD=${2:-"error"}
COUNT=0

if [ ! -f "$LOGFILE" ]; then
    echo "File not found!"
    exit 1
fi

while IFS= read -r LINE; do
    if echo "$LINE" | grep -iq "$KEYWORD"; then
        COUNT=$((COUNT + 1))
    fi
done < "$LOGFILE"

echo "Keyword '$KEYWORD' found $COUNT times"

echo "Last 5 matching lines:"
grep -i "$KEYWORD" "$LOGFILE" | tail -5
```



Script 5 — The Open Source Manifesto Generator (Unit 5)

Starter structure

```
#!/bin/bash
# Script 5: Open Source Manifesto
Generator

echo "Answer the following questions:"

read -p "1. Tool you use daily: " TOOL
read -p "2. What does freedom mean to
you? " FREEDOM
read -p "3. What will you build? " BUILD

DATE=$(date '+%d %B %Y')
OUTPUT="manifesto_$(whoami).txt"

echo "On $DATE, I believe in open
source." > $OUTPUT
echo "I use $TOOL daily." >> $OUTPUT
echo "Freedom means $FREEDOM to
me." >> $OUTPUT
echo "I want to build $BUILD and share
it with the world." >> $OUTPUT

echo "Manifesto saved to $OUTPUT"
cat $OUTPUT
```

```
ishan@Ishani MINGW64 ~/OneDrive/Documents/OSS_Project
$ nano script5.sh

ishan@Ishani MINGW64 ~/OneDrive/Documents/OSS_Project
$ ./script5.sh
Answer the following questions:
1. Tool you use daily: Vs
2. What does freedom mean to you? Life
3. What will you build? Bright Future
Manifesto saved to manifesto_ishan.txt
On 30 March 2026, I believe in open source.
I use Vs daily.
Freedom means Life to me.
I want to build Bright Future and share it with the world.
```

Conclusion

This project gave me a clear understanding of how open-source systems actually work in practice. I got familiar with the core ideas behind open-source philosophy, explored the structure of the Linux system, and learned the basics of shell scripting.

Working with tools like Git also helped me see how real-world development happens. It's not just about writing code individually, but about collaborating, managing changes, and building something together.

Overall, I realized that Git is more than just a tool—it plays a key role in modern software development by enabling collaboration, transparency, and continuous innovation.

References

- GNU Official Website
- Git Documentation
- Linux Manual Pages